

# CatLink

## The Smart EV Grid Powered by Nokia Network APIs

Unifying EV charging networks with 5G for real-time connectivity, identity security, and AI optimisation.

TALENT ARENA 2026 - OPEN GATEWAY HACKATHON



Made with GAMMA

## The Real Problem

# 30% of EV drivers abandon charging sessions due to unreliable infrastructure

Lack of infrastructure is breaking the promise of electric mobility — and costing operators millions.

*"Maria has 15% battery left. She opens 3 different apps. Drives 12 minutes to a charger. It's offline."*



### Fragmented Networks

Repsol, Iberdrola, Endesa, Tesla, Ionity — each with separate accounts and authentication



### Outdated Status

Chargers shown as available are often offline, with no real-time demand balancing



### Security Gaps

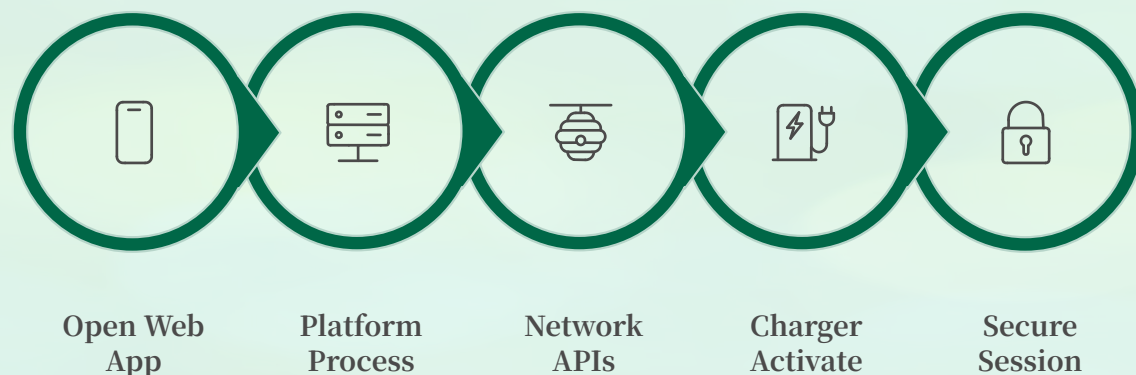
Inconsistent payment systems and weak fraud prevention expose SIM swap risks

**⚠ The impact: Lost time for users → Lost revenue for operators → Slower EV adoption across Europe**

⚡ THE SOLUTION

# CatLink: The First Unified Smart EV Grid Powered by Network as Code

CatLink connects all EV chargers into one intelligent, secure, real-time network using 5G APIs and agentic AI.



One platform. Every operator. Real-time intelligence at every step of the charging journey.



Made with GAMMA



## User Experience

# A Seamless & Secure EV Charging Experience

Designed from the driver's perspective — no more juggling apps, guessing availability, or worrying about fraud.



### One Unified App

Access every operator through a single interface



### Real-Time Availability

Live charger status — no more wasted journeys



### Instant Verification

Identity confirmed and fraud blocked before the session starts



### AI-Optimised Routing

Recommended chargers based on location, demand density, network status, and predicted congestion

From fragmentation to a unified intelligent grid.

## Technical Architecture

# Purposeful Use of Network APIs & Agentic AI

Every API call serves a clear function across the charging journey. An AI agent orchestrates and makes operational decisions.



### Before Charging

**Number Verification API** — instant identity check

**SIM Swap API** — fraud risk detection

**Location Verification API** — confirms user is at charger



### During Charging

**Device Status API** — real-time charger monitoring

**Quality on Demand API** — prioritised connectivity for auth, payment, and session stability



### Intelligence Layer

**Agentic AI** (Gemini + MCP) orchestrates API calls dynamically, balances demand with density data, predicts congestion, and reroutes users — all within cost and rate limits



AGENTIC AI USAGE



COMBINED NETWORK APIS

Financial Impact

# Quantified ROI for Operators

Conservative scenario for a mid-size city with **5,000 chargers** and **100,000 sessions per day**.

€96K

Lost Daily

8% session failure rate = 8,000 failed sessions/day

€14M

Recovered Annually

CatLink reduces failures by 40%, recovering  
€38.4K/day

€18M+

Total Annual Impact

Including €4M in fraud prevention savings

## Business Model

01

SaaS per Charger

€4–6/month subscription

02

Per-Session Fee

Security verification charge

03

Data Intelligence

Analytics subscription tier

04

Revenue Share

Performance-based model

# CatLink Demo

**The Smart EV Grid Powered by Nokia Network APIs**